

SHAHIN RAHBARIASL

Senior Software Engineer · Distributed Systems · AdTech

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SUMMARY

Senior Software Engineer with 9+ years designing and scaling distributed systems in mission-critical production environments. Deep expertise in Go, architecting services that handle **600B+ daily requests** with sub-10ms latency across 12 global data centers. Proven track record in AdTech infrastructure, real-time data pipelines, and open-source contributions. M.Sc. Computer Science, University of Waterloo — published researcher at SIGIR.

TECHNICAL SKILLS

Languages	Go · Python · Java · TypeScript · JavaScript · C++ · C# · Bash
Systems & Data	Kafka · Spark · Redis · Cassandra · Aerospike · PostgreSQL · MySQL · MongoDB · Elasticsearch · MariaDB
Cloud & DevOps	AWS (S3, Lambda, Kinesis, Athena, Glue) · GCP · IBM Cloud · Kubernetes · Docker · Ansible · ArgoCD · OPA
Observability	Grafana · InfluxDB · Prometheus · Vault · Zookeeper
Web & APIs	gRPC · GraphQL · Hasura · REST · React · Node.js · Angular · Express · Bootstrap

EXPERIENCE

Index Exchange

Senior Software Engineer

March 2023 – Present

- Architect core ad exchange services processing **~600B requests/day**, fanned out to 100+ demand partners generating **~1.7T downstream requests daily**.
- Maintain **sub-10ms end-to-end latency** across concurrent Go services on thousands of machines in **12 global data centers**; decisions made within single-digit milliseconds.
- Owning the **full redesign and rebuild of the company-wide authorization system** — one of the most critical infrastructure projects — using OPA, Kubernetes, ArgoCD, GraphQL, Hasura, and Go.
- Building and maintaining **external and internal APIs** that power customer-facing dashboards and internal tooling across the organization.
- Built Privacy Sandbox exchange components and header bidding solutions across Web, CTV, and Mobile; open-source contributor to **Prebid** and **Adscert** (IAB security protocol).

Seneca College — Software Design & Data Science

Faculty (Part-time)

Jan – April 2024

- Developed and delivered Web Programming Principles curriculum; mentored students on modern frameworks and tech career development.

Index Exchange

Software Engineer II

May 2021 – March 2023

- Designed high-performance Go services and APIs for the core ad exchange platform with distributed low-latency architecture.
- Built storage layer (Aerospike, MariaDB, MySQL), streaming pipelines (Kafka), and full observability stack (Grafana, InfluxDB).
- Automated CI/CD workflows with Ansible and GitHub/GitLab across large-scale global infrastructure.

IBM Cloud and Cognitive Software

Data Engineer

March 2020 – May 2021

- Built core IBM Cloud services: authentication controllers, policy management, and Commerce back-end serving thousands of enterprise customers.
- Architected telemetry data pipelines for cloud infrastructure monitoring at scale; contributed during pandemic-era digital transformation surge.

Blockthrough

July 2018 – March 2020

Full Stack Software Engineer

- Joined as engineer #10 and scaled with team to **50+ engineers** — owned data pipeline architecture, real-time monitoring, and backend infrastructure.
- Built **Bidthru** (Prebid wrapper) processing ~1B auctions/day, generating **\$1M+ revenue** in its first iteration.
- Developed publisher analytics dashboards and animation detection tooling for ad inventory quality assurance.

BMO Capital Markets

May – August 2017

Software Developer (Internship)

- Built real-time client/trader analytics platform used by trading floor managers for P&L performance tracking.
- Optimized financial databases and implemented live trade monitoring for risk management and compliance.

University of Waterloo

2016 – 2018

Research Assistant — IR & Data Systems Group

- Conducted research in information retrieval and HCI under Dr. Mark D. Smucker; led user studies and evaluation tooling; published 2 papers at SIGIR 2019.

EDUCATION

University of Waterloo

2016 – 2018

M.Sc. in Computer Science

Iran University of Science and Technology

2012 – 2016

B.Sc. in Computer Science

SELECTED PUBLICATIONS

SIGIR 2019	Time-Limits and Summaries for Faster Relevance Assessing
SIGIR 2019	Efficient Construction of Test Collections Using Dynamic Sampling
Thesis, 2019	The Effects of Time Constraints and Document Excerpts on Relevance Assessing Behaviour — M.Sc. Thesis, University of Waterloo
Book, 2015	Web Development Using Drupal CMS (ISBN: 9789648340419)

OPEN SOURCE CONTRIBUTIONS

- **Adscert** (IAB Tech Lab, 2021) — Cryptographic authentication protocol for securing programmatic ad transactions — now an industry standard.
- **Prebid** (2021) — Leading open-source header bidding library enabling publishers worldwide to maximize ad revenue.

AWARDS & RECOGNITION

- Blockthrough Employee of the Year 2018–19
- International Masters Student Award, University of Waterloo 2016
- Graduate Research Studentship, University of Waterloo 2016
- Top 1% — Iran's National University Entrance Exam (300,000+ participants) 2012
- Full Scholarship — Iran University of Science and Technology 2012